Anomaly Detection on the Edge for Measurement and Control in Distributed Energy Resources Network

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PROBLEM STATEMENT

- Measurement and Control signals are communicated using **Sunspec Modbus , DNP3 AN2018-001, IEC 61850-7-420.**
- Increased Risk of **Cyber Attacks** such as MITM, DOS, Remote Access from both the measurement and control signals.
- Lack of **Anomaly Detection System** to monitor DER communication.

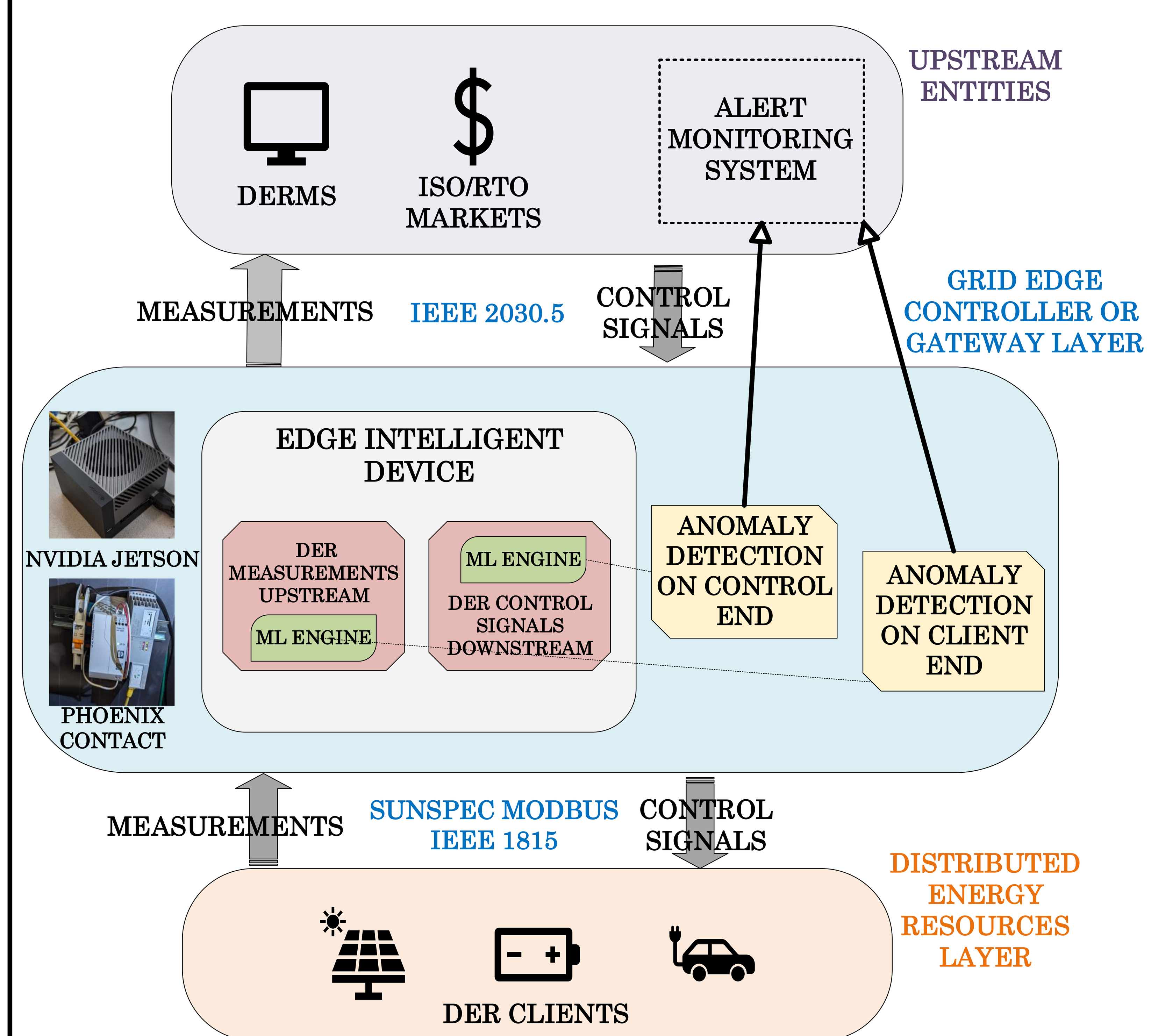
EXISTING WORK

- Anomaly Detection System (ADS) that **monitors the measurement telemetry** from the DER clients to the Edge Intelligent Devices (EIDs).
- [1, 2] has a Decision Tree based Machine Learning ADS (**ML-ADS**) that parses the network traffic from the DER clients to the EID and detects anomalies.
- ***Limitations:*** The **control signals** from the upstream entities (DERMS, etc) to the EIDs are **not monitored**.

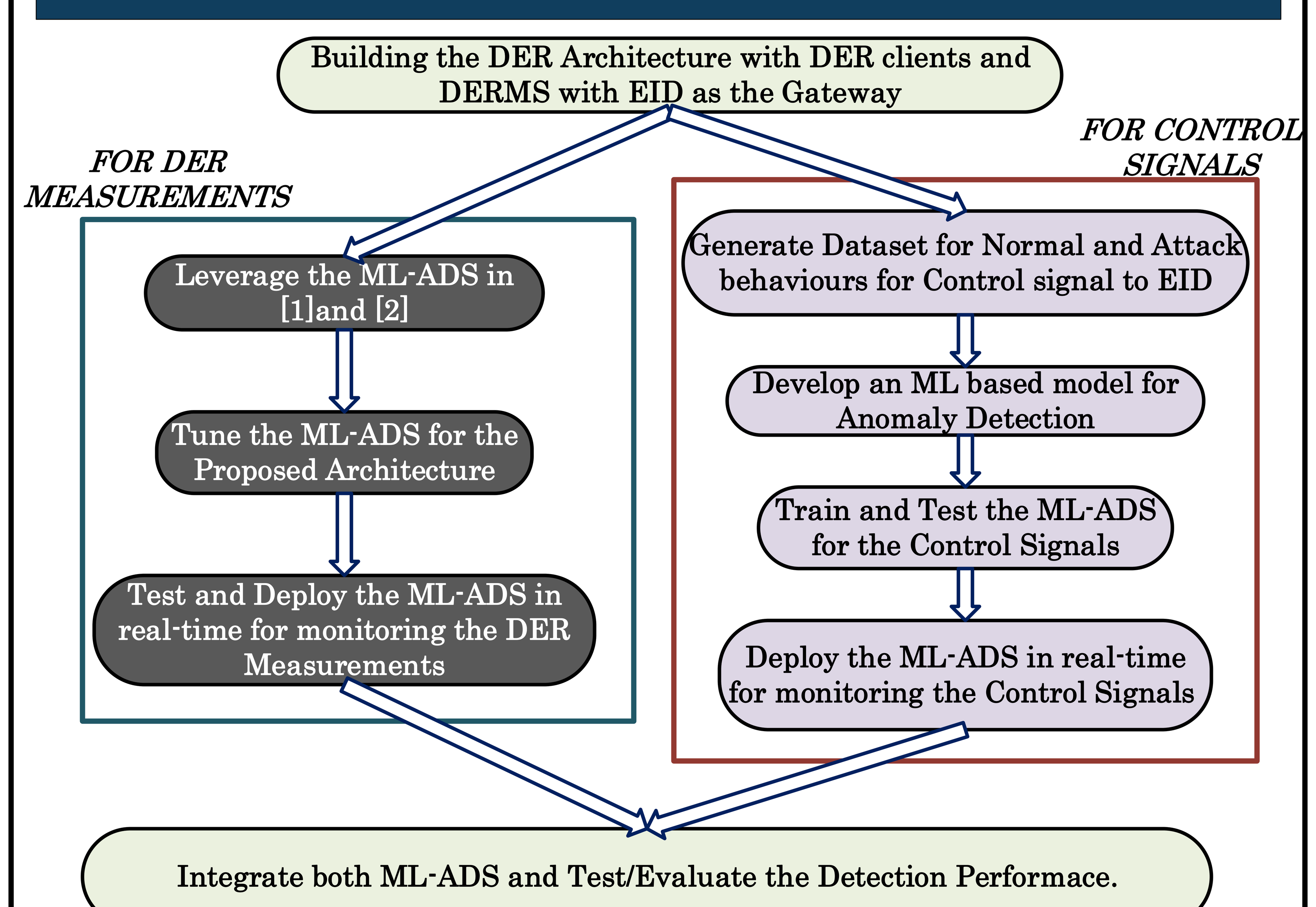
PROPOSED SOLUTION

- Design and Develop Machine Learning based Anomaly Detection System (ML-ADS) for **monitors the control signals** from the upstream entities (DERMS, etc) to the EID.
- Train the ML- ADS for identifying various cyber-attacks.
- Deploy the trained ML-ADS to detect anomalies.

PROPOSED ARCHITECTURE



METHODOLOGY



IMPLEMENTATION

- Building a baseline **DER network** and test DER operation such as polling DER measurement and send control signals.
- Designing the best performing ML Algorithm for detecting the anomalies in the control signals.
- Deploy the ML-ADS in real-time along with the baseline DER network traffic.

EXPECTED OUTCOME

- The **Integrated Model** will monitor the anomalies both on the DER Measurements and the control signals.
- The integrated model is expected to have **better Accuracy** for anomaly detection with **lower Latency**.

REFERENCES

[1] M. Abdelkhalek and M. Govindarasu, "ML-based Anomaly Detection System for DER DNP3 Communication in Smart Grid," 2022 IEEE International Conference on Cyber Security and Resilience (CSR), Rhodes, Greece, 2022, pp. 209-214, doi: 10.1109/CSR54599.2022.9850313.

[2] I. Siniosoglou, P. Radoglou-Grammatikis, G. Efstathopoulos, P. Fouliras and P. Sarigiannidis, "A Unified Deep Learning Anomaly Detection and Classification Approach for Smart Grid Environments," in IEEE Transactions on Network and Service Management, vol. 18, no. 2, pp. 1137-1151, June 2021, doi: 10.1109/TNSM.2021.3078381

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